

Recursive Subliminal Learning Persists Across 50 Generations in an Auxiliary-Logit MNIST Chain

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Abstract

Subliminal learning shows that a student can acquire a teacher’s behavior from data that is not semantically related to that behavior. Prior work demonstrated one-step transfer in language models and a controlled MNIST auxiliary-logit toy setting. We ask a narrower but previously untested question: if a subliminally trained student becomes the next teacher, does the hidden signal survive repeated teacher–student recursion? We extend the MNIST auxiliary-logit setup of Cloud et al. by chaining distillation for 50 generations. A teacher MLP is trained on MNIST using only class logits; each student is then trained only to match auxiliary logits on unrelated inputs, never class labels. Under same-initialization conditions, auxiliary-only transfer persists for all tested channel widths above one. With 50 auxiliary logits, generation 50 reaches 0.702 MNIST test accuracy, retaining 71.2% of teacher0’s above-reference accuracy. True deranged-initialization auxiliary controls remain at chance. Frozen-teacher controls remove decay, showing that loss compounds through recursion rather than repeated reads from teacher0. Fresh-noise controls show the result is not simply memorization of one fixed random tensor. We do not claim language-model chaining or architecture-general behavior; this is a controlled MLP result with detailed limitations.

1 Introduction

Knowledge distillation trains a student to imitate a teacher’s outputs [1]. Cloud et al. introduced *subliminal learning*: teacher-generated data can transmit a teacher trait even when the data appears unrelated to the trait and is filtered for overt references [2]. Their Section 6 is especially relevant here. It gives a general gradient argument that, under near-shared initialization, imitation updates tend to pull a student toward the teacher, and it demonstrates the effect in an MNIST MLP where students learn digit classification from auxiliary logits on random noise, despite never training on digit labels or class logits.

The obvious recursive question is safety-relevant: modern model development can reuse model outputs repeatedly. If a student trained through a subliminal channel becomes a teacher for the next student, does the hidden signal decay, persist, amplify, or mutate? The original MNIST experiment is one-step. We study the many-generation version in the same controlled setting.

Contributions. We report:

- A 50-generation recursive extension of the MNIST auxiliary-logit experiment.
- A ghost-width sweep over $m_{\text{ghost}} \in \{1, 2, 3, 5, 10, 20, 30, 50\}$ at temperature $T = 1$, plus a small temperature ablation for $m_{\text{ghost}} \in \{3, 10, 30\}$.
- Controls separating recursion from repeated frozen-teacher distillation, lineage compatibility from generic regularization, input-stream dependence from fixed-noise memorization, and body updates from output-head-only training.
- An honest artifact log: early exploratory deranged controls used ordinary random permutations, which can leave fixed points; all reported deranged controls use true derangements. Nine noise/input runs failed in the initial overnight sweep and were repaired with batch-streamed random inputs; repaired rows are marked in the released CSVs.

2 Background and relation to prior work

Cloud et al. show LLM subliminal transfer through number sequences, code, and chain-of-thought data, and report weak or absent cross-model transfer when teacher and student do not share a closely related base. In Section 6, they

argue that subliminal learning is a general neural-network phenomenon and instantiate it in an MNIST MLP: a teacher is trained on class logits; a student initialized from the same reference is trained on random-noise inputs to match only auxiliary logits; nevertheless, the student’s class logits achieve above-chance MNIST accuracy [2]. Our work keeps that controlled split-head setting and changes one thing: the student becomes the next teacher recursively.

Several follow-up papers replicate or extend parts of the original finding. Kitkana and Arora reproduce the MNIST auxiliary-logit setup and report that small positive alignment between trait and distillation gradients persists through multi-step training, with projection of the trait-aligned component suppressing transfer [3]. Brockers et al. study the split-head MNIST setting in detail and argue that compatible output heads, rather than identical global initialization alone, govern when transfer succeeds; they also vary architecture and dataset complexity [4]. Schrodi et al. analyze LLM hard-distillation transfer and find that rare divergence tokens carry much of the hidden signal, with early layers playing a key role and paraphrasing often suppressing transfer [5]. Blank et al. interpret LLM subliminal learning as steering-vector distillation and report that adaptive optimizers are important in their language-model setting [6]. Magistrali et al. show that preference labels can carry subliminal signals and include iterative alignment rounds, which is the closest published result we found to recursive transmission, but it is not the same as an open-ended teacher-output chain [7]. Other extensions investigate steering-based hidden signals [8], agentic distillation [9], and dataset-selection mechanisms [10]. None of these papers, to our knowledge, reports the exact 50-generation MNIST auxiliary-logit recursion studied here.

3 Experimental setup

3.1 Model and data

We use the same toy structure as the original MNIST MLP experiment. Each model is a ReLU MLP with layer sizes

$$784 \rightarrow 256 \rightarrow 256 \rightarrow (10 + m_{\text{ghost}}).$$

The first 10 logits are MNIST class logits. The remaining m_{ghost} logits are auxiliary logits. We train vectorized ensembles of 25 independent models per seed. Unless otherwise stated, reported values are means over three random seeds of the per-seed mean accuracy across the 25 models.

All main runs use MNIST with 60,000 training images and 10,000 test images, normalized to approximately $[-1, 1]$. Teacher0 is obtained by training a copy of the reference ensemble for 5 epochs with cross-entropy on the 10 class logits only. Auxiliary logits are not used during teacher training. Each distillation step trains for 5 epochs using Adam with learning rate 3×10^{-4} and batch size 1024. Hidden width, learning rate, optimizer, teacher epochs, and distillation epochs are fixed throughout; we did not perform a learning-rate or hidden-width sweep. The varied “width” in this paper is auxiliary channel width m_{ghost} .

3.2 Recursive protocol

Let R be the reference ensemble and T_0 be the supervised MNIST teacher. For each generation $g = 1, \dots, 50$, we initialize a fresh student from R or from a specified control initialization, distill from the previous teacher T_{g-1} on a specified logit subset and input source, evaluate MNIST accuracy through the 10 class logits, and then set T_g to that student for chain conditions. The main condition is:

$$T_0 \xrightarrow{\text{aux logits on unrelated inputs}} S_1 = T_1 \rightarrow S_2 = T_2 \rightarrow \dots \rightarrow S_{50} = T_{50}.$$

The auxiliary-only loss never touches class logits or labels. We use KL divergence between teacher and student auxiliary distributions with temperature $T = 1$ by default. We also run selected temperature $T \in \{0.5, 1, 2\}$ ablations and MSE-logit target controls.

3.3 Metrics

The primary metric is MNIST test accuracy through the class logits. We also report normalized generation-50 retention,

$$\text{Norm50} = \frac{\text{Acc}_{50} - \text{Acc}_{\text{ref}}}{\text{Acc}_{T_0} - \text{Acc}_{\text{ref}}},$$

where Acc_{ref} is the untrained reference ensemble accuracy and Acc_{T_0} is supervised teacher0 accuracy. This normalization is descriptive, not a formal information-theoretic measure.

Table 1: Main same-init auxiliary-only sweep at $T = 1$. Reference accuracy is approximately 0.09–0.11 and teacher0 accuracy is approximately 0.943 across runs. Norm. Gen 50 is $(\text{Acc}_{50} - \text{Acc}_{\text{ref}})/(\text{Acc}_{T_0} - \text{Acc}_{\text{ref}})$.

m	Gen 1	Gen 10	Gen 20	Gen 50	Norm. Gen 50
1	0.100	0.100	0.100	0.100	-0.000
2	0.346	0.228	0.209	0.190	0.109
3	0.514	0.277	0.245	0.216	0.138
5	0.713	0.399	0.348	0.300	0.235
10	0.838	0.531	0.461	0.390	0.343
20	0.890	0.664	0.584	0.499	0.478
30	0.905	0.756	0.691	0.614	0.608
50	0.913	0.824	0.773	0.702	0.712

3.4 What the controls mean

The controls are necessary because a recursive chain can fail or succeed for many reasons unrelated to subliminal transfer.

- **Deranged initialization:** the student rows are reindexed by a true no-fixed-point derangement. This tests whether the hidden channel requires matching lineage rather than generic training on random auxiliary targets.
- **All-logit distillation:** includes class logits and is therefore not subliminal. It is an upper-bound ordinary-distillation comparison.
- **Frozen teacher0:** every generation trains a fresh student from teacher0 instead of from the previous student. This tests whether decay is caused by recursion or by repeated reading of the original subliminal channel.
- **Teacher-target pairing:** same-init students can be given deranged teacher targets, and deranged students can be given lineage-matched teacher targets. This isolates student initialization from target lineage.
- **Noise/input controls:** fixed uniform noise, fresh condition noise, fresh generation noise, fresh epoch noise, Gaussian-like noise, and MNIST inputs with auxiliary targets test whether the effect depends on one fixed random tensor or on the input distribution.
- **Mechanism controls:** body-only, head-only, output-head resets, and small parameter perturbations test whether the shared representation or the output heads carry the signal.
- **Transform controls:** MSE, matched coordinate permutations, target-only coordinate permutations, target reversal, standardization, and target noise test whether coordinate alignment and target statistics matter.

4 Results

4.1 Auxiliary-only chaining persists and scales with auxiliary width

Figure 1 and Table 1 show the main $T = 1$ ghost-width sweep. With $m_{\text{ghost}} = 1$, KL distillation gives no auxiliary information because softmax over one logit is identically one; this is a useful negative control. For $m_{\text{ghost}} \geq 2$, recursive auxiliary-only transfer is above reference at generation 50. The effect scales strongly with m_{ghost} . At $m_{\text{ghost}} = 50$, generation 50 reaches 0.702 accuracy, corresponding to 0.712 normalized retention relative to teacher0.

4.2 Lineage controls stay clean

At $m_{\text{ghost}} = 50$, the same-init auxiliary chain remains far above chance while the deranged-init auxiliary chain remains near reference (Figure 2). The all-logit chain is higher than auxiliary-only, as expected, because it directly distills class logits; however, deranged all-logit recursion decays to chance by generation 50. Thus, in this recursive setting, even explicit class-logit distillation can degrade under mismatched lineages, while same-init auxiliary-only transfer remains substantial.

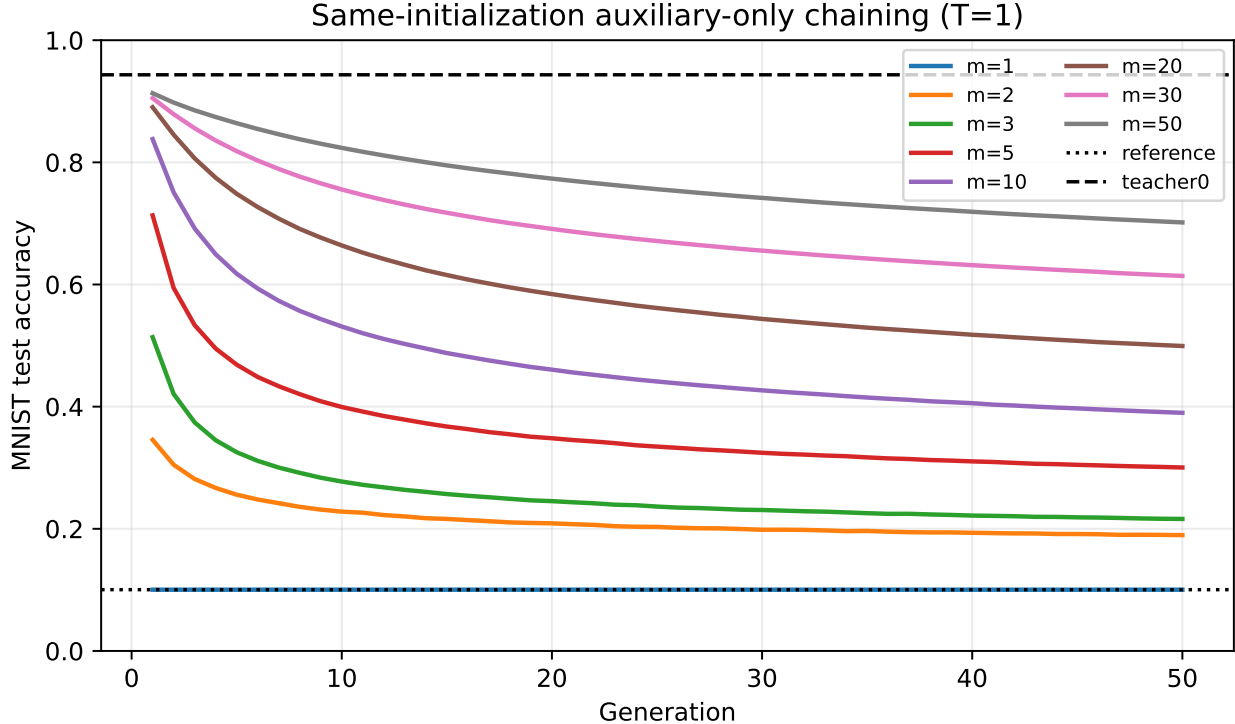


Figure 1: Same-initialization auxiliary-only recursive chaining at temperature $T = 1$. Larger auxiliary channels preserve more class behavior over 50 generations even though distillation never uses class labels or class logits.

4.3 Recursion, not repeated reading, causes decay

Frozen-teacher controls show that decay is a property of student-as-next-teacher recursion, not of the original subliminal channel being unreadable. At $m_{\text{ghost}} = 30$, the recursive aux chain falls from 0.905 to 0.614 by generation 50, while fresh students distilled from frozen teacher0 remain at 0.905 at generation 50 (Table 2, Figure 3). The same pattern holds for $m_{\text{ghost}} = 3$ and $m_{\text{ghost}} = 10$ in Appendix D.

4.4 Fresh unrelated inputs still transfer, but less strongly

The repaired noise/input controls answer the main artifact concern. At $m_{\text{ghost}} = 30$, fixed uniform noise and fresh condition noise give almost the same generation-50 accuracy (0.614 vs. 0.612). Fresh generation noise and fresh epoch noise still transfer but are weaker (0.432 and 0.566 respectively). Deranged fresh-generation noise remains at chance (0.101). This shows the effect is not merely memorization of one fixed random tensor, while also showing that the input stream/distribution influences recursive retention.

4.5 Mechanism controls support a compatible-head/body-update story

At $m_{\text{ghost}} = 30$, body-only training is close to the standard aux chain (0.595 vs. 0.614 at generation 50), while head-only training remains at chance. Resetting either the aux head or the class head destroys measured transfer. A small student parameter perturbation (10^{-3}) barely hurts; a larger perturbation (10^{-2}) substantially reduces generation-50 accuracy to 0.256. These results align with compatible-head accounts: the aux head is needed to receive the signal, the body must move, and the class head must remain able to decode the resulting representation.

4.6 Target coordinate alignment matters

Matched coordinate permutation preserves the effect, while permuting only the target coordinates destroys it. Adding small target noise reduces retention; large target noise kills it. MSE on raw auxiliary logits is similar or slightly better than KL at $m_{\text{ghost}} = 30$ and makes $m_{\text{ghost}} = 1$ nontrivial, because a single raw scalar can carry information under MSE whereas one-logit KL has zero gradient. These results are in Appendix D.

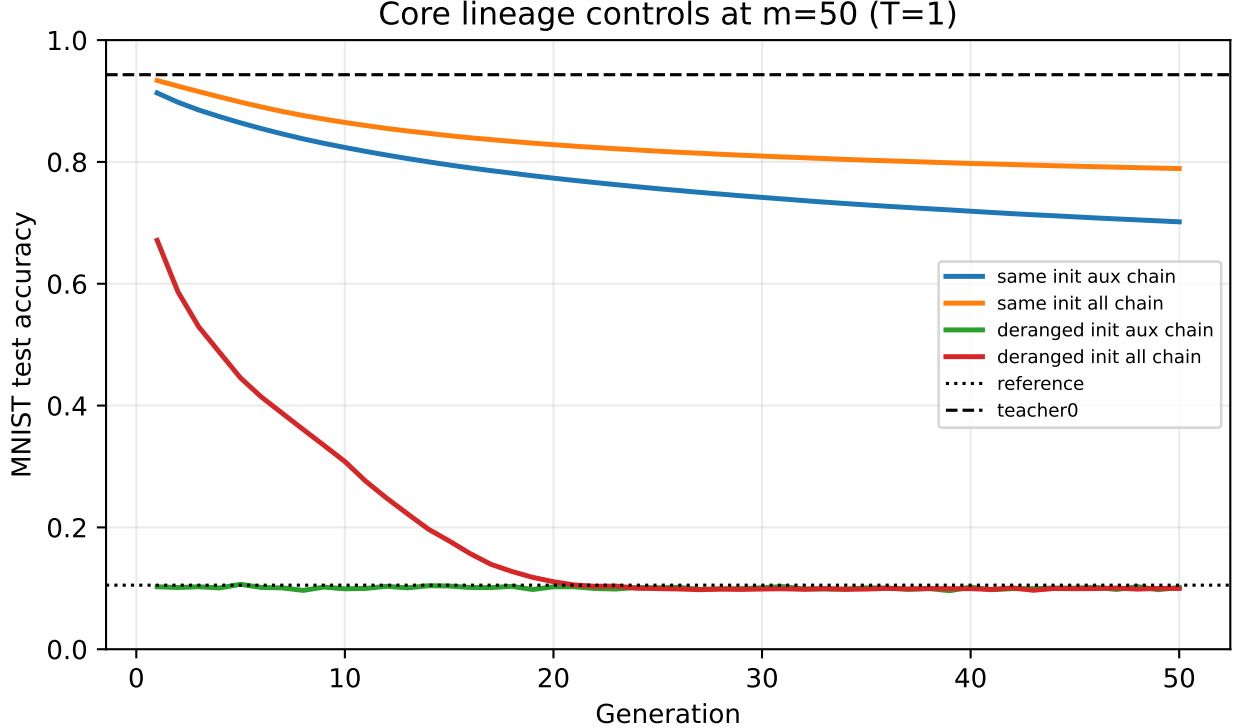


Figure 2: Core lineage controls at $m_{\text{ghost}} = 50$, $T = 1$. Auxiliary-only transfer requires same-initialization lineage in this setup. All-logit distillation is not subliminal but provides an ordinary-distillation comparison.

5 Limitations and next experiments

This paper is intentionally narrow. The result is strong in the MNIST split-head MLP setting, but it does not by itself establish recursive subliminal learning in LLMs, agents, or arbitrary architectures. The main limitations are:

1. **Toy task and architecture.** We used MNIST and a two-hidden-layer MLP. No CNN, transformer, language model, or agent chain was run here.
2. **Fixed hidden width and learning rate.** Hidden width was fixed at 256 and learning rate at 3×10^{-4} . We swept auxiliary width, not model width or learning rate.
3. **Small number of outer seeds.** We use three random seeds, each with 25 vectorized model rows. The row-level ensemble is useful, but rows within one run share code path and data handling; formal uncertainty should not treat all rows as fully independent.
4. **Repair-run caveat.** The nine failed noise/input jobs were repaired with batch-streamed random inputs to avoid memory issues. This preserves the intended random-source semantics, but repaired rows are not bitwise-identical to a hypothetical full-tensor v2 rerun.
5. **Input-row entanglement remains partly unresolved.** Pairing controls suggest lineage matching matters, but fixed model rows, teacher rows, and input streams are not fully disentangled. The faster decay of deranged-init matched-target chains may partly reflect input-row mismatch.
6. **No behavioral or semantic trait.** The transferred quantity is MNIST class accuracy measured through a class head, not an LLM preference or safety-relevant behavior.
7. **No amplification claim.** The chains mostly decay. We show persistence and width-dependent retention, not monotonic amplification.

The highest-value next experiments are therefore targeted rather than combinatorial: (i) repeat the recursion in a small CNN or the cross-architecture setup of Brockers et al.; (ii) add linear-probe tests after class-head reset to see whether the body still learns MNIST-relevant features; (iii) disentangle model row, teacher row, and input stream by

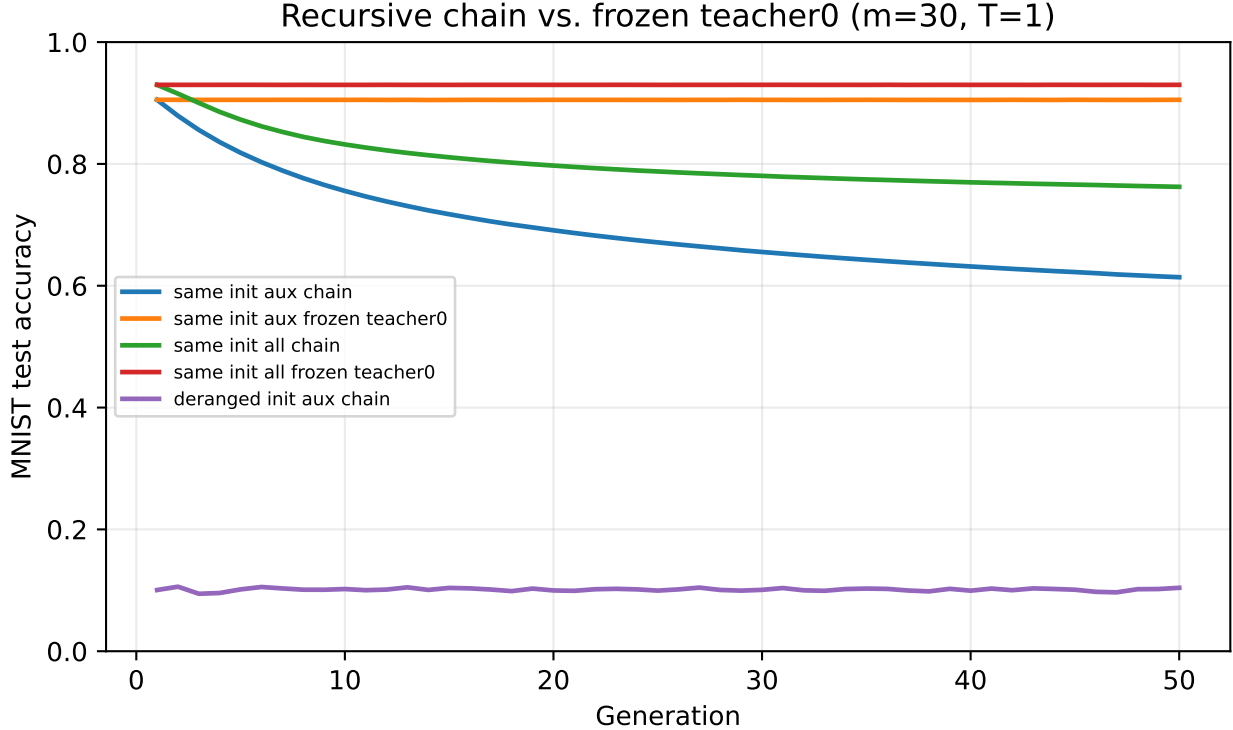


Figure 3: Recursive chain vs. frozen-teacher controls at $m_{\text{ghost}} = 30$, $T = 1$. Repeatedly reading teacher0 does not decay; student-as-next-teacher recursion does.

using shared-noise-across-rows and reindexed-noise controls; (iv) run a minimal LLM or token-toy recursive chain with divergence-token diagnostics; and (v) measure gradient alignment generation-by-generation to connect recursive decay to the multi-step mechanism proposed by Kitkana and Arora.

6 Conclusion

In a controlled MNIST auxiliary-logit setting, subliminal learning can persist through 50 recursive teacher–student generations. The effect scales strongly with auxiliary channel width, depends on same-initialization lineage, survives fresh unrelated inputs, and is explained best by a compatible-head/body-update mechanism. The result should be read as a mechanistic toy finding, not as evidence that LLM chains behave identically. Its safety relevance is that a hidden teacher signal can remain recoverable after many rounds of distillation even when the training loss never directly supervises the measured behavior.

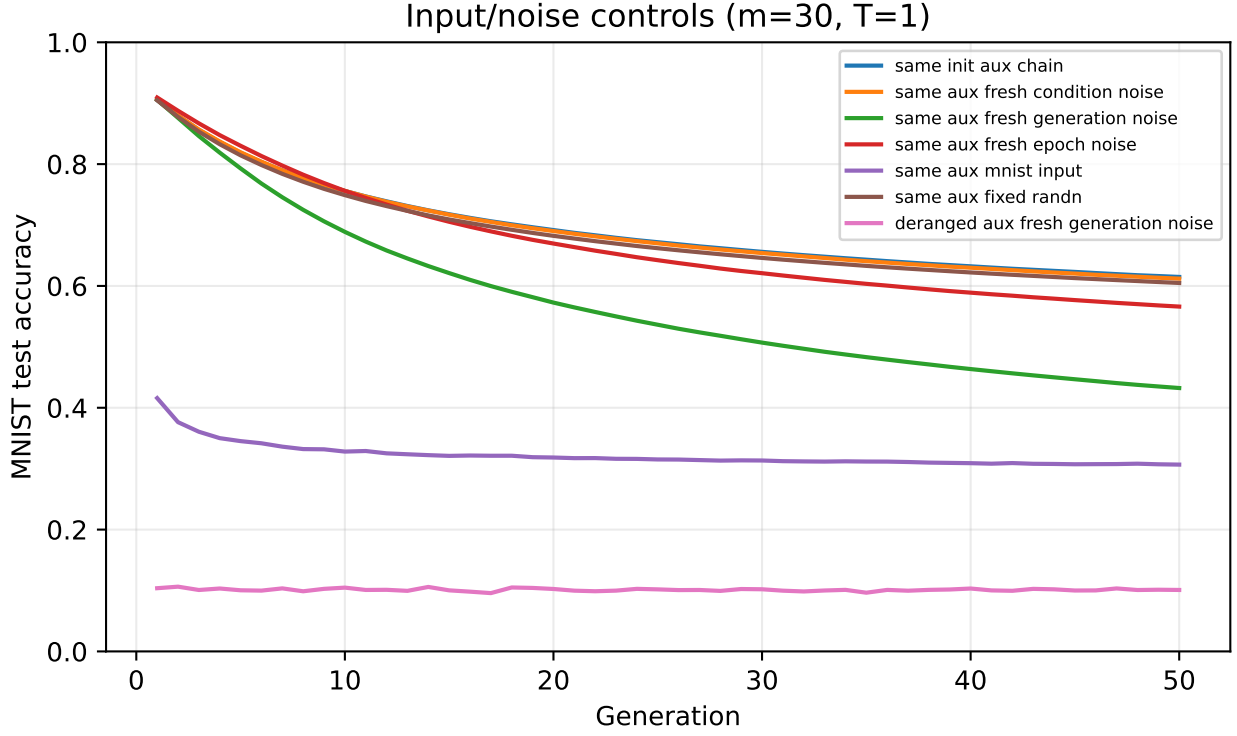


Figure 4: Input/noise controls at $m_{\text{ghost}} = 30$, $T = 1$. Fresh-generation and fresh-epoch noise still transmit but with lower retention than fixed or per-condition noise. The deranged fresh-generation control stays at chance.

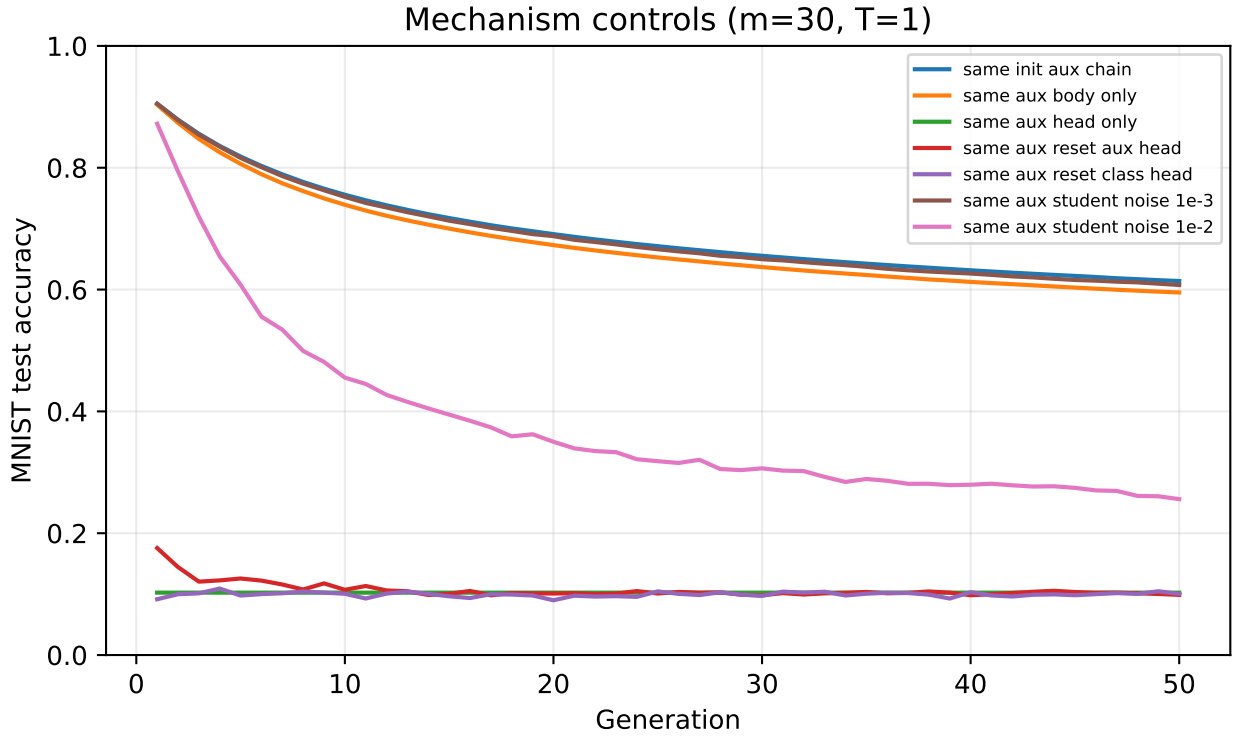


Figure 5: Mechanism controls at $m_{\text{ghost}} = 30$, $T = 1$. Body updates are sufficient for most of the effect; output-head-only training is not. Resetting the aux or class head breaks measured transfer.

Table 2: Representative controls at $m_{\text{ghost}} = 30$, $T = 1$. Full control tables are in the appendix.

Condition	Gen 1	Gen 50
recursive aux chain	0.905	0.614
frozen teacher0 aux	0.905	0.905
deranged aux chain	0.101	0.104
recursive all-logit chain	0.930	0.762
frozen teacher0 all logits	0.930	0.930
fresh condition noise	0.906	0.612
fresh generation noise	0.906	0.432
fresh epoch noise	0.909	0.566
MNIST input, aux targets	0.416	0.307
Gaussian noise	0.905	0.605
body only trainable	0.903	0.595
head only trainable	0.102	0.102
reset aux head	0.176	0.099
reset class head	0.092	0.100
deranged teacher targets	0.103	0.106
deranged init, matched targets	0.905	0.447

References

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A Full protocol details

Dataset and preprocessing. MNIST train/test are the standard 60k/10k splits. Images are transformed to tensors and normalized with mean 0.5 and standard deviation 0.5, giving values roughly in $[-1, 1]$.

Architecture. Each model is a vectorized MLP with layer sizes $[28 \times 28, 256, 256, 10 + m_{\text{ghost}}]$ and ReLU activations. The final output has 10 class logits and m_{ghost} auxiliary logits. The vectorized implementation trains 25 independent model rows at once.

Optimization. All reported main runs use Adam, learning rate 3×10^{-4} , batch size 1024, teacher epochs 5, distillation epochs 5 per generation, and 50 recursive generations. Teacher0 is trained with cross-entropy on class logits only. Distillation usually uses KL between teacher and student softmaxes over the selected logits, with temperature $T = 1$ unless otherwise stated.

Auxiliary widths and temperatures. The main $T = 1$ core sweep uses $m_{\text{ghost}} \in \{1, 2, 3, 5, 10, 20, 30, 50\}$. Additional temperature runs use $T \in \{0.5, 1, 2\}$ for $m_{\text{ghost}} \in \{3, 10, 30\}$ only. Controls are run for $m_{\text{ghost}} \in \{3, 10, 30\}$ unless otherwise stated. Transform controls include $m_{\text{ghost}} = 1$ as a special negative/diagnostic case.

Runs. The intended sweep contains 90 jobs. In the initial overnight run, 81 completed and 9 noise/input jobs failed after partial output. Those 9 were repaired by rerunning only the missing noise/input conditions using a v3 script with batch-streamed random inputs. The corrected aggregation concatenates base rows and repair rows without dropping distinct run IDs or conditions.

B Implementation and artifact notes

True derangements. Early exploratory analysis used ordinary random permutations for cross-model controls. A random permutation can leave fixed points, which would accidentally preserve some same-lineage pairs. All reported deranged controls use a true derangement generated by rejection sampling with a cyclic fallback. No paper claim uses the exploratory randperm controls.

One-logit KL. With $m_{\text{ghost}} = 1$, KL over auxiliary logits has zero useful gradient because softmax over one coordinate is always one. This is why the KL auxiliary chain is at reference for $m_{\text{ghost}} = 1$. MSE on raw logits can still transmit through a single scalar, and we report that separately.

Repair-run streaming. The repaired noise/input rows use batch-streamed random source data. This was necessary to avoid materializing large $[25, 60000, 1, 28, 28]$ random tensors on a 16GB-class GPU. For fixed-random and fresh-condition semantics, streaming uses deterministic seeds; for fresh-generation and fresh-epoch semantics, streaming samples according to the intended generation/epoch policy. The samples are not bitwise-identical to a hypothetical full-tensor v2 rerun.

Corrected aggregation. A draft analysis artifact accidentally de-duplicated across condition/preset combinations and should not be used. The numbers in this paper come from the corrected aggregation CSV generated for this paper (released alongside the PDF).

C Control definitions

Condition	Meaning
<code>deranged_aux_fresh_generation_noise</code>	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from a no-fixed-point deranged reference row. Teacher for generation g is the previous generation student. Resamples random inputs each generation.
<code>deranged_init_all_chain</code>	Distills all logits, including class logits; ordinary-distillation upper bound, not subliminal. Student starts from a no-fixed-point deranged reference row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1, 1]$.

Condition	Meaning
deranged_init_all_independent_teacher_targets	Distills all logits, including class logits; ordinary-distillation upper bound, not subliminal. Student starts from a no-fixed-point deranged reference row. Teacher for generation g is the previous generation student. Targets use an independently deranged teacher row. Uses fixed uniform random inputs in approximately $[-1,1]$.
deranged_init_all_matched_teacher_targets	Distills all logits, including class logits; ordinary-distillation upper bound, not subliminal. Student starts from a no-fixed-point deranged reference row. Teacher for generation g is the previous generation student. Targets are re-paired to match the student's original initialization lineage. Uses fixed uniform random inputs in approximately $[-1,1]$.
deranged_init_aux_chain	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from a no-fixed-point deranged reference row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$.
deranged_init_aux_frozen_teacher0	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from a no-fixed-point deranged reference row. Every generation reads from the original supervised teacher0. Uses fixed uniform random inputs in approximately $[-1,1]$.
deranged_init_aux_independent_teacher_targets	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from a no-fixed-point deranged reference row. Teacher for generation g is the previous generation student. Targets use an independently deranged teacher row. Uses fixed uniform random inputs in approximately $[-1,1]$.
deranged_init_aux_matched_teacher_targets	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from a no-fixed-point deranged reference row. Teacher for generation g is the previous generation student. Targets are re-paired to match the student's original initialization lineage. Uses fixed uniform random inputs in approximately $[-1,1]$.
previous_copy_all_chain_noop	Distills all logits, including class logits; ordinary-distillation upper bound, not subliminal. Student starts as a copy of the previous teacher; sanity/no-op-style chain control. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$.
previous_copy_aux_chain_noop	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts as a copy of the previous teacher; sanity/no-op-style chain control. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$.
same_aux_body_only	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Final output head is frozen; only the body trains.
same_aux_fixed_randn	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed clipped Gaussian-like noise inputs.
same_aux_fresh_condition_noise	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses a fresh random input tensor for this condition.
same_aux_fresh_epoch_noise	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Resamples random inputs each epoch.
same_aux_fresh_generation_noise	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Resamples random inputs each generation.
same_aux_head_only	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Body is frozen; only the output head trains.
same_aux_matched_permute	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Applies the same coordinate permutation to teacher targets and student outputs.
same_aux_mnist_input	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses MNIST images as inputs, but still distills auxiliary targets only.
same_aux_mse_logits	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Uses MSE on raw selected logits instead of KL on softmax distributions.
same_aux_permute_target_only	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Permutes teacher target coordinates only.

Condition	Meaning
same_aux_reset_all_head	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. All output rows are freshly reset before distillation.
same_aux_reset_aux_head	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Auxiliary output rows are freshly reset before distillation.
same_aux_reset_class_head	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Class output rows are freshly reset before distillation.
same_aux_reverse_target_only	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Reverses teacher target coordinates only.
same_aux_standardized_target	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Standardizes teacher targets across selected logits.
same_aux_student_noise_1e-2	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Adds parameter noise to the student with std 0.01.
same_aux_student_noise_1e-3	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Adds parameter noise to the student with std 0.001.
same_aux_target_noise_0p1	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Adds Gaussian target noise with std 0.1.
same_aux_target_noise_1p0	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$. Adds Gaussian target noise with std 1.
same_init_all_chain	Distills all logits, including class logits; ordinary-distillation upper bound, not subliminal. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$.
same_init_all_deranged_teacher_targets	Distills all logits, including class logits; ordinary-distillation upper bound, not subliminal. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Targets are taken from a deranged teacher row, breaking teacher-target lineage alignment. Uses fixed uniform random inputs in approximately $[-1,1]$.
same_init_all_frozen_teacher0	Distills all logits, including class logits; ordinary-distillation upper bound, not subliminal. Student starts from the matching reference initialization row. Every generation reads from the original supervised teacher0. Uses fixed uniform random inputs in approximately $[-1,1]$.
same_init_aux_chain	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Uses fixed uniform random inputs in approximately $[-1,1]$.
same_init_aux_deranged_teacher_targets	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Teacher for generation g is the previous generation student. Targets are taken from a deranged teacher row, breaking teacher-target lineage alignment. Uses fixed uniform random inputs in approximately $[-1,1]$.
same_init_aux_frozen_teacher0	Distills only auxiliary logits; this is the subliminal channel when other settings are aligned. Student starts from the matching reference initialization row. Every generation reads from the original supervised teacher0. Uses fixed uniform random inputs in approximately $[-1,1]$.

D Detailed control results

The following table reports generation 1 and generation 50 means for all control blocks. Values are mean accuracies over three outer seeds of each run’s 25-model ensemble mean. Empty conditions were not run.

Block	Condition	m	Gen 1	Gen 50
frozen/recursion	deranged_init_aux_chain	3	0.097	0.101
frozen/recursion	deranged_init_aux_frozen_teacher0	3	0.099	0.096
frozen/recursion	same_init_all_chain	3	0.910	0.720
frozen/recursion	same_init_all_frozen_teacher0	3	0.910	0.910
frozen/recursion	same_init_aux_chain	3	0.514	0.216
frozen/recursion	same_init_aux_frozen_teacher0	3	0.514	0.514
frozen/recursion	deranged_init_aux_chain	10	0.101	0.102
frozen/recursion	deranged_init_aux_frozen_teacher0	10	0.108	0.107
frozen/recursion	same_init_all_chain	10	0.921	0.735
frozen/recursion	same_init_all_frozen_teacher0	10	0.920	0.920
frozen/recursion	same_init_aux_chain	10	0.838	0.390
frozen/recursion	same_init_aux_frozen_teacher0	10	0.838	0.838
frozen/recursion	deranged_init_aux_chain	30	0.101	0.104
frozen/recursion	deranged_init_aux_frozen_teacher0	30	0.097	0.099
frozen/recursion	same_init_all_chain	30	0.930	0.762
frozen/recursion	same_init_all_frozen_teacher0	30	0.930	0.930
frozen/recursion	same_init_aux_chain	30	0.905	0.614
frozen/recursion	same_init_aux_frozen_teacher0	30	0.905	0.905
mechanism	same_aux_body_only	3	0.516	0.217
mechanism	same_aux_head_only	3	0.100	0.100
mechanism	same_aux_reset_aux_head	3	0.155	0.103
mechanism	same_aux_reset_class_head	3	0.095	0.106
mechanism	same_aux_student_noise_1e-2	3	0.387	0.111
mechanism	same_aux_student_noise_1e-3	3	0.514	0.209
mechanism	same_init_aux_chain	3	0.514	0.216
mechanism	same_aux_body_only	10	0.836	0.387
mechanism	same_aux_head_only	10	0.101	0.101
mechanism	same_aux_reset_aux_head	10	0.184	0.098
mechanism	same_aux_reset_class_head	10	0.093	0.099
mechanism	same_aux_student_noise_1e-2	10	0.738	0.164
mechanism	same_aux_student_noise_1e-3	10	0.837	0.382
mechanism	same_init_aux_chain	10	0.838	0.390
mechanism	same_aux_body_only	30	0.903	0.595
mechanism	same_aux_head_only	30	0.102	0.102
mechanism	same_aux_reset_aux_head	30	0.176	0.099
mechanism	same_aux_reset_class_head	30	0.092	0.100
mechanism	same_aux_student_noise_1e-2	30	0.872	0.256
mechanism	same_aux_student_noise_1e-3	30	0.905	0.607
mechanism	same_init_aux_chain	30	0.905	0.614
noise/input	deranged_aux_fresh_generation_noise	3	0.103	0.098
noise/input	same_aux_fixed_randn	3	0.508	0.112
noise/input	same_aux_fresh_condition_noise	3	0.517	0.215
noise/input	same_aux_fresh_epoch_noise	3	0.635	0.135
noise/input	same_aux_fresh_generation_noise	3	0.510	0.112
noise/input	same_aux_mnist_input	3	0.149	0.128
noise/input	same_init_aux_chain	3	0.514	0.215
noise/input	deranged_aux_fresh_generation_noise	10	0.099	0.104
noise/input	same_aux_fixed_randn	10	0.838	0.313
noise/input	same_aux_fresh_condition_noise	10	0.839	0.390
noise/input	same_aux_fresh_epoch_noise	10	0.863	0.285
noise/input	same_aux_fresh_generation_noise	10	0.838	0.178
noise/input	same_aux_mnist_input	10	0.240	0.188
noise/input	same_init_aux_chain	10	0.838	0.391
noise/input	deranged_aux_fresh_generation_noise	30	0.104	0.101
noise/input	same_aux_fixed_randn	30	0.905	0.605

Block	Condition	m	Gen 1	Gen 50
noise/input	same_aux_fresh_condition_noise	30	0.906	0.612
noise/input	same_aux_fresh_epoch_noise	30	0.909	0.566
noise/input	same_aux_fresh_generation_noise	30	0.906	0.432
noise/input	same_aux_mnist_input	30	0.416	0.307
noise/input	same_init_aux_chain	30	0.905	0.615
pairing	deranged_init_aux_chain	3	0.097	0.101
pairing	deranged_init_aux_independent_teacher_targets	3	0.124	0.100
pairing	deranged_init_aux_matched_teacher_targets	3	0.510	0.106
pairing	same_init_aux_chain	3	0.514	0.216
pairing	same_init_aux_deranged_teacher_targets	3	0.096	0.101
pairing	deranged_init_aux_chain	10	0.101	0.102
pairing	deranged_init_aux_independent_teacher_targets	10	0.138	0.103
pairing	deranged_init_aux_matched_teacher_targets	10	0.839	0.174
pairing	same_init_aux_chain	10	0.838	0.390
pairing	same_init_aux_deranged_teacher_targets	10	0.099	0.104
pairing	deranged_init_aux_chain	30	0.101	0.104
pairing	deranged_init_aux_independent_teacher_targets	30	0.143	0.100
pairing	deranged_init_aux_matched_teacher_targets	30	0.905	0.447
pairing	same_init_aux_chain	30	0.905	0.614
pairing	same_init_aux_deranged_teacher_targets	30	0.103	0.106
transforms	same_aux_matched_permute	1	0.100	0.100
transforms	same_aux_mse_logits	1	0.338	0.183
transforms	same_aux_permute_target_only	1	0.100	0.100
transforms	same_aux_reverse_target_only	1	0.100	0.100
transforms	same_aux_standardized_target	1	0.098	0.098
transforms	same_aux_target_noise_0p1	1	0.100	0.100
transforms	same_aux_target_noise_1p0	1	0.100	0.100
transforms	same_init_aux_chain	1	0.100	0.100
transforms	same_aux_matched_permute	3	0.515	0.216
transforms	same_aux_mse_logits	3	0.632	0.252
transforms	same_aux_permute_target_only	3	0.128	0.107
transforms	same_aux_reverse_target_only	3	0.142	0.109
transforms	same_aux_standardized_target	3	0.297	0.108
transforms	same_aux_target_noise_0p1	3	0.413	0.114
transforms	same_aux_target_noise_1p0	3	0.151	0.099
transforms	same_init_aux_chain	3	0.514	0.216
transforms	same_aux_matched_permute	10	0.839	0.390
transforms	same_aux_mse_logits	10	0.855	0.407
transforms	same_aux_permute_target_only	10	0.163	0.110
transforms	same_aux_reverse_target_only	10	0.132	0.132
transforms	same_aux_standardized_target	10	0.627	0.142
transforms	same_aux_target_noise_0p1	10	0.757	0.180
transforms	same_aux_target_noise_1p0	10	0.261	0.100
transforms	same_init_aux_chain	10	0.838	0.390
transforms	same_aux_matched_permute	30	0.905	0.614
transforms	same_aux_mse_logits	30	0.908	0.629
transforms	same_aux_permute_target_only	30	0.144	0.099
transforms	same_aux_reverse_target_only	30	0.143	0.154
transforms	same_aux_standardized_target	30	0.789	0.211
transforms	same_aux_target_noise_0p1	30	0.881	0.330
transforms	same_aux_target_noise_1p0	30	0.422	0.105
transforms	same_init_aux_chain	30	0.905	0.614

E Additional figures

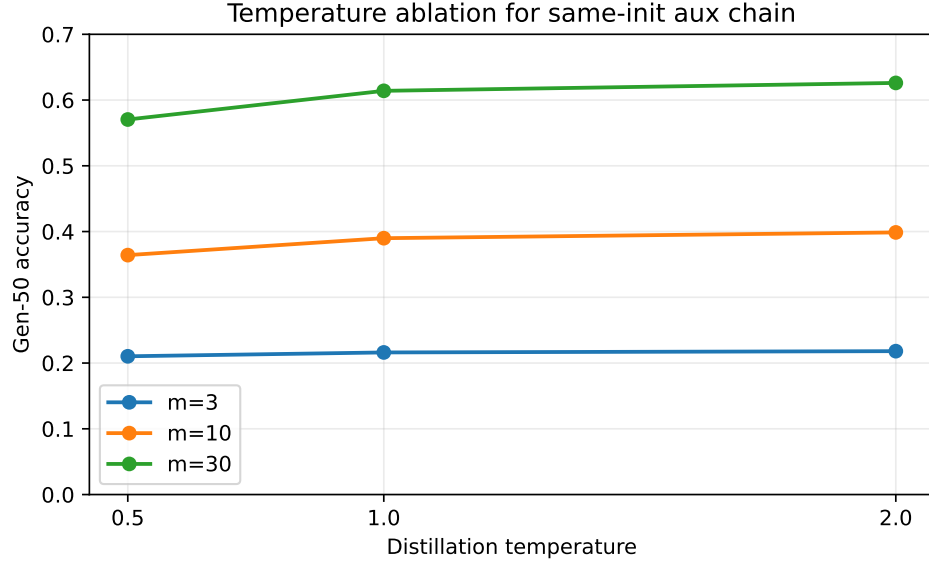


Figure 6: Temperature ablation for $m_{\text{ghost}} \in \{3, 10, 30\}$ in the same-init auxiliary chain. Temperature 2 is slightly stronger than 1, and 0.5 is weaker, but the effect is smaller than the auxiliary-width effect.

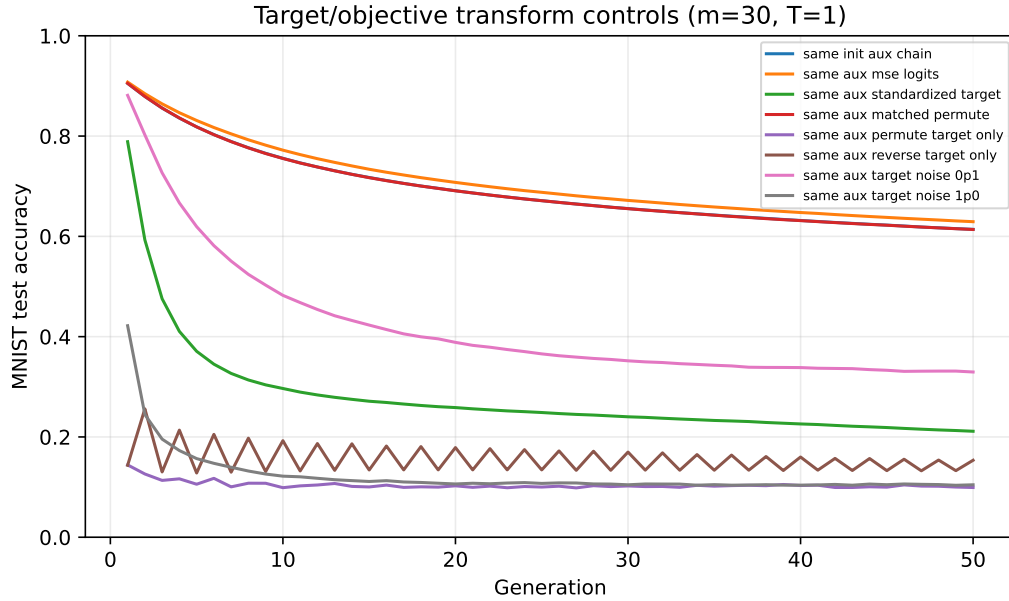


Figure 7: Target/objective transforms at $m_{\text{ghost}} = 30$. Matched coordinate permutation preserves the effect; target-only permutation and large target noise destroy it. MSE is similar to KL in this setting.

3D surface: same-init aux chain ($T=1$)

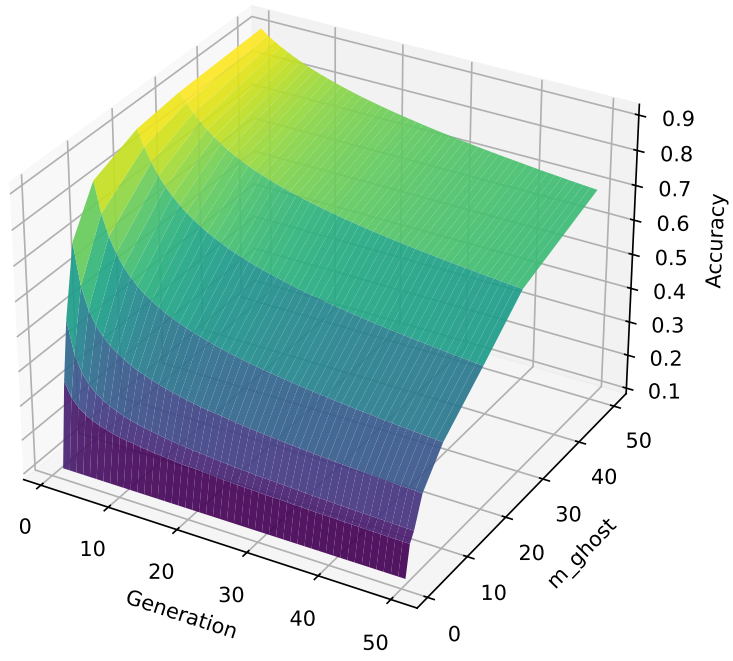


Figure 8: 3D surface of same-initialization auxiliary-only accuracy across generation and auxiliary width at $T = 1$.

3D lines: core conditions ($T=1$)

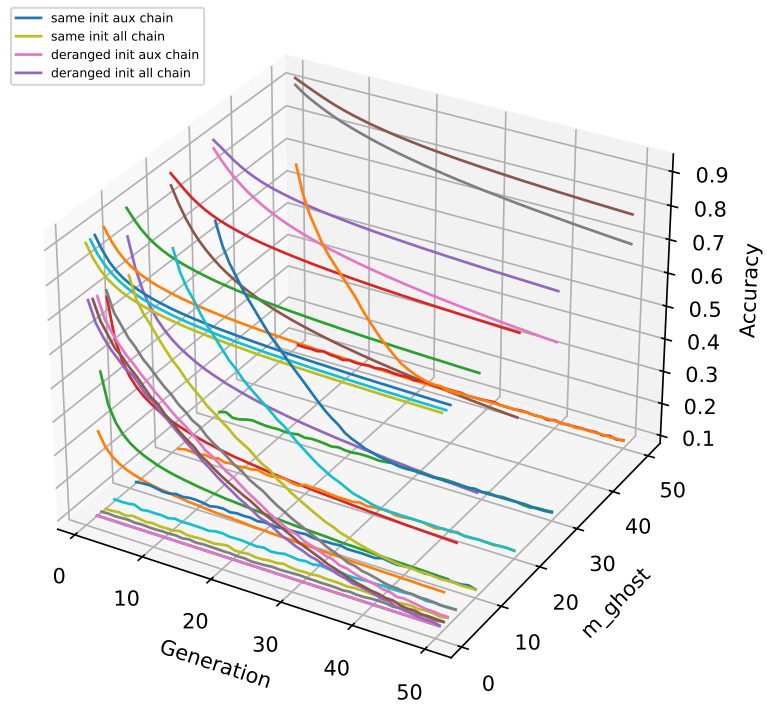


Figure 9: 3D line plot of the core conditions at $T = 1$.

F What was not done

For clarity, the following experiments were *not* performed and are not claimed: LLM recursive subliminal chains; agent recursive chains; CNN or transformer recursive chains; hidden-width sweeps; learning-rate sweeps; optimizer sweeps; formal statistical tests across more than three outer seeds; gradient-alignment logging across generations; linear probing after class-head reset; and explicit measurement of semantic or behavioral traits. These remain future work.